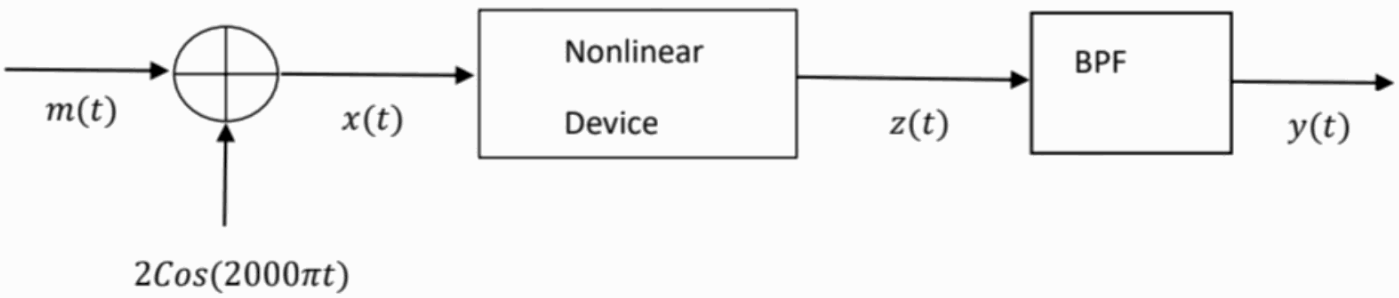
The message signal $m(t) = 5Cos(200\pi t) + 10Cos(400\pi t)$ is transmitted by using the DSB-Large Carrier (LC)-AM modulator given below. The nonlinear device has the input-output relationship $z(t) = 60x(t) + x^2(t)$. The bandpass filter (BPF) has a gain of 1, $BW = 2B = 400\ Hz$, center frequency $f_C = 1\ kHz$.



a) Plot $m(t), x(t),\ z(t),\ y(t)$.

b) Find and plot $M(f),\ X(f),\ Z(f),\ Y(f)$. (Do the filtering in frequency domain. Do not use filter command!)

c) Demodulate $y(t)$ by using envelope detector. Plot the demodulated signal $\tilde{m}(t)$ and its spectra. Compare your result to the original message signal $m(t)$. (You can use envelope command of MATLAB. However, if you use the circuit of the envelope detector to demodulate, you get extra points.)

$(x+y+z)^2 = x^2+y^2+z^2+2(xy+xz+yz)$

a) $m(t) = 5\cos(200\pi t) + 10\cos(400\pi t)$

$T = 0{,}01$

at 0 $m(t) = 5+10 = 15$

at 0,01 $m(t) = 5\cos(2\pi) + 10\cos(4\pi)$

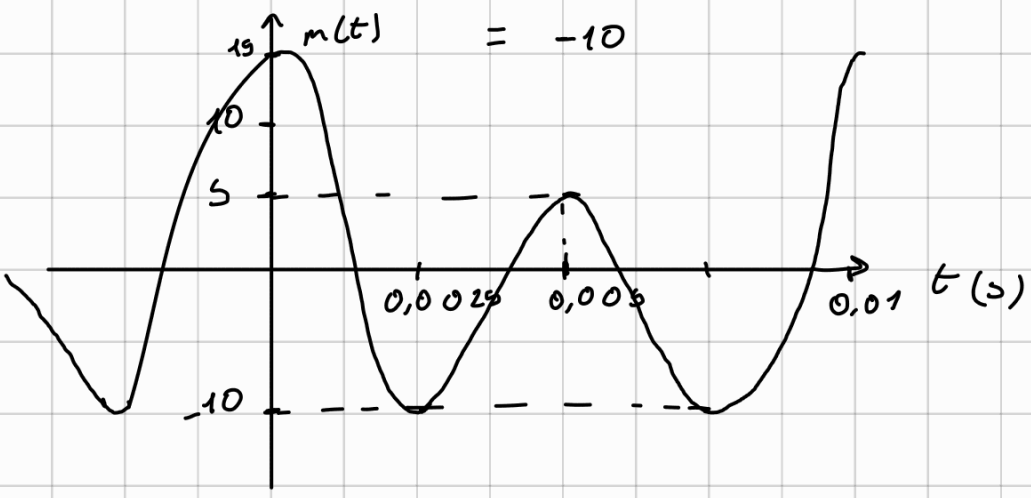
$= 15$

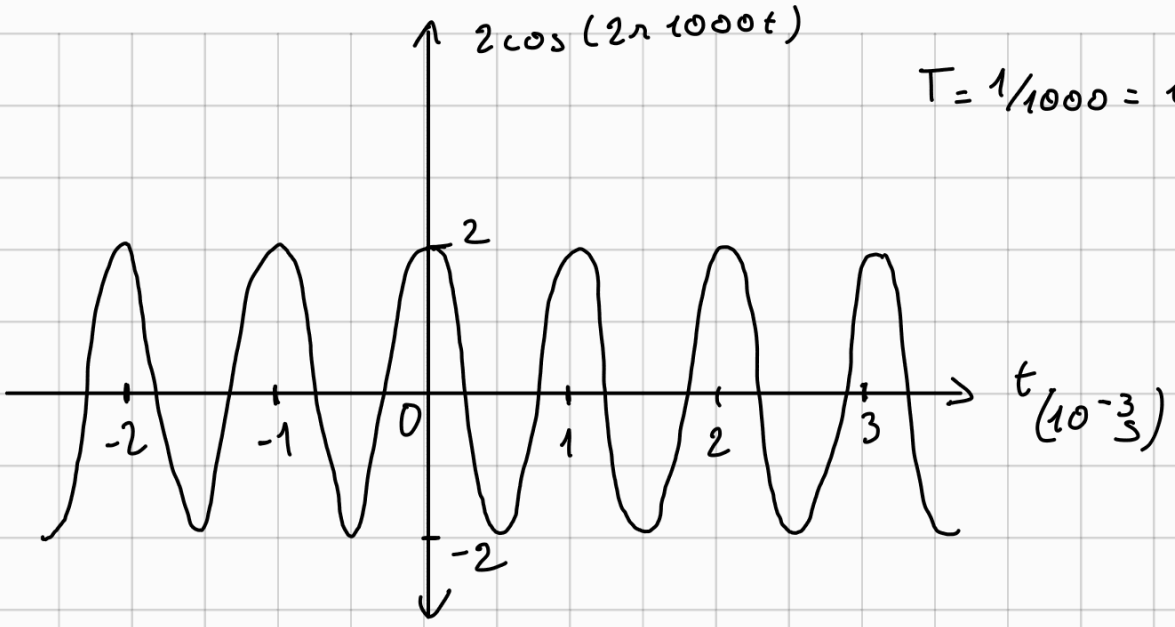
at 0,005 $m(t) = 5\cos(\pi) + 10\cos(2\pi)$

$= -5+10 = 5$

at 0,0025 $m(t) = 5\cos(\pi/2) + 10\cos(\pi)$

$= -10$





$T = 1/1000 = 10^{-3}$

$x(t) = 5\cos(200\pi t) + 10\cos(400\pi t) + 2\cos(2000\pi t)$

$z(t) = 60\,x(t) + x^2(t)$

$= 60\left[5\cos(200\pi t) + 10\cos(400\pi t) + 2\cos(2000\pi t)\right]$
$+ \left(5\cos(200\pi t)\right)^2 + \left(10\cos(400\pi t)\right)^2 + \left(2\cos(2000\pi t)\right)^2$
$+ 2\left[\left(5\cos(200\pi t)\cdot 10\cos(400\pi t)\right) + \left(5\cos(200\pi t)\cdot 2\cos(2000\pi t)\right) + \left(10\cos(400\pi t)\cdot 2\cos(2000\pi t)\right)\right]$

$= 300\cos(2\pi 100t) + 600\cos(2\pi 200t) + 120\cos(2\pi 1000t)$
$+ 25\cos^2(2\pi 100t) + 100\cos^2(2\pi 200t) + 4\cos^2(2\pi 1000t)$
$+ 100\cos(2\pi 100t)\cdot\cos(2\pi 200t) + 20\cos(2\pi 100t)\cos(2\pi 1000t)$
$+ 40\cos(2\pi 200t)\cos(2\pi 1000t)$

$\cos^2(\alpha) = \frac{1}{2}(1 + \cos(2\alpha))$

$\cos x \cdot \cos y = \frac{1}{2}\left[\cos(x+y) + \cos(x-y)\right]$

$$z(t) = 300\cos(2\pi 100t) + 600\cos(2\pi 200t) + 120\cos(2\pi 1000t)$$
$$+ \frac{25}{2} + \frac{25}{2}\cos(2\pi 200t) + 50 + 50\cos(2\pi 400t) + 2 + 2\cos(2\pi 2000t)$$
$$+ 50\cos(2\pi 300t) + 50\cos(2\pi 100t) + 10\cos(2\pi 1100t) + 10\cos(2\pi 900t)$$
$$+ 20\cos(2\pi 1200t) + 20\cos(2\pi 800t)$$

$$z(t) = \frac{129}{2} + 350\cos(2\pi 100t) + \frac{1225}{2}\cos(2\pi 200t)$$
$$+ 50\cos(2\pi 300t) + 50\cos(2\pi 400t) + 20\cos(2\pi 800t)$$
$$+ 10\cos(2\pi 900t) + 120\cos(2\pi 1000t) + 10\cos(2\pi 1100t)$$
$$+ 20\cos(2\pi 1200t) + 2\cos(2\pi 2000t)$$

$$BW = 2B = 400\ hz \quad , f_c = 1\ khz$$

$$y(t) = 20\cos(2\pi 800t) + 10\cos(2\pi 900t) + 120\cos(2\pi 1000t)$$
$$+ 10\cos(2\pi 1100t) + 20\cos(2\pi 1200t)$$

b) $m(t) = 5\cos(200\pi t) + 10\cos(400\pi t)$

$= 5\cos(2\pi 100t) + 10\cos(2\pi 200t)$

$M(f) = \frac{5}{2}\left[\delta(f+100t) + \delta(f-100t)\right] + 5\left[\delta(f+200) + \delta(f-200)\right]$

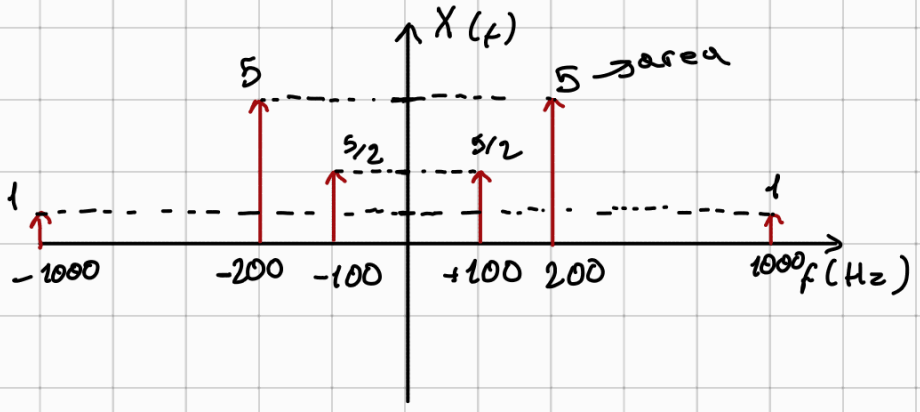


$x(t) = m(t) + 2\cos(2\pi 1000t)$

$X(f) = M(f) + \delta(f+1000t) + \delta(f-1000t)$

$= \frac{5}{2}\left[\delta(f+100t) + \delta(f-100t)\right] + 5\left[\delta(f+200) + \delta(f-200)\right]$

$+ \delta(f+1000t) + \delta(f-1000t)$



$z(t) = \frac{129}{2} + 350\cos(2\pi 100t) + \frac{1225}{2}\cos(2\pi 200t)$

$+ 50\cos(2\pi 300t) + 50\cos(2\pi 400t) + 20\cos(2\pi 800t)$

$+ 10\cos(2\pi 900t) + 120\cos(2\pi 1000t) + 10\cos(2\pi 1100t)$

$+ 20\cos(2\pi 1200t) + 2\cos(2\pi 2000t)$

$\Rightarrow Z(f) = \frac{129}{2}\delta(f) + 175\left(\delta(f+100) + \delta(f-100)\right)$ ($\frac{129}{2} = 64.5$)

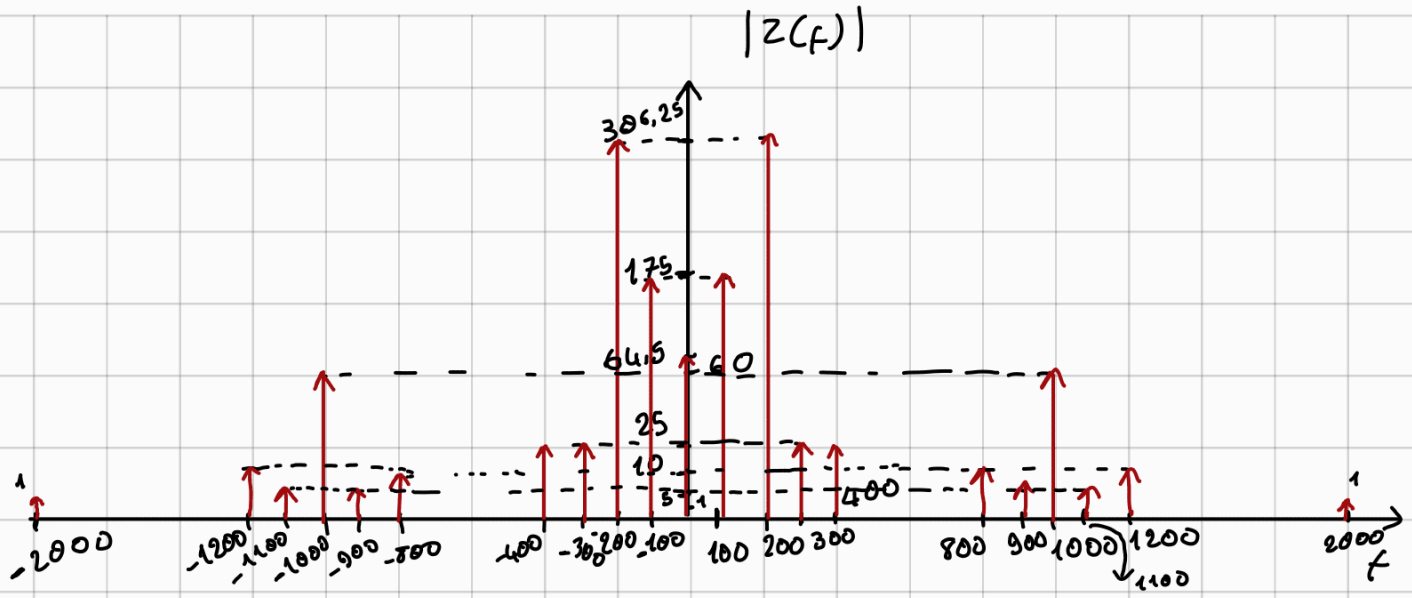
$+ 306{,}25\left(\delta(f+200) + \delta(f-200)\right) + 25\left(\delta(f+300) + \delta(f-300)\right)$

$+ 25\left(\delta(f+400) + \delta(f-400)\right) + 10\left(\delta(f+800) + \delta(f-800)\right)$

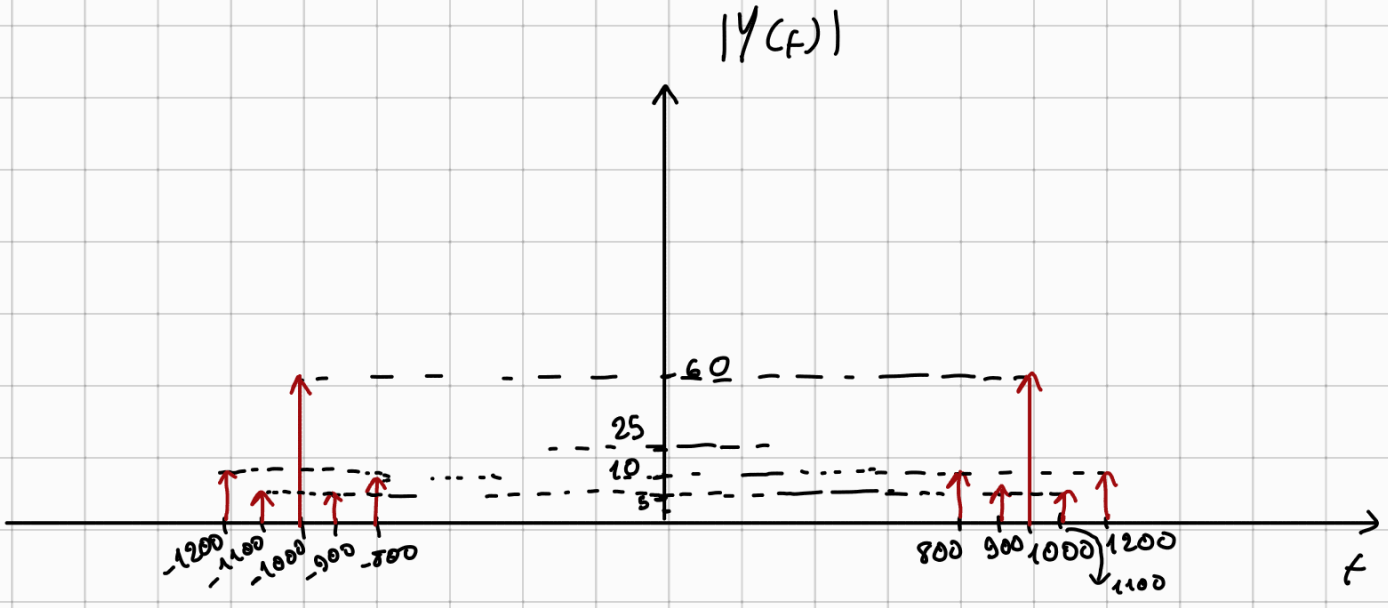
$+ 5\left(\delta(f+900) + \delta(f-900)\right) + 60\left(\delta(f+1000) + \delta(f-1000)\right)$

$+ 5\left(\delta(f+1100) + \delta(f-1100)\right) + 10\left(\delta(f+1200) + \delta(f-1200)\right)$

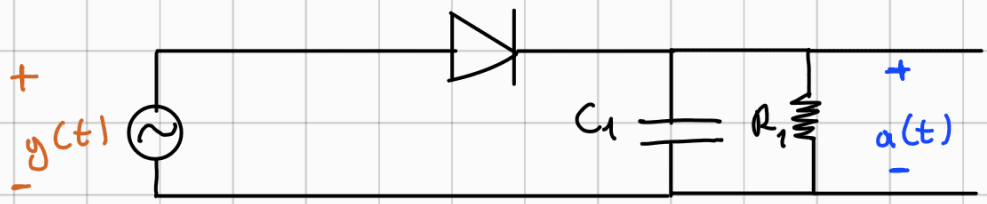
$+ \left(\delta(f+2000) + \delta(f-2000)\right)$ //



$$\Rightarrow Y(f) = 10(\delta(f+800)+\delta(f-800))$$
$$+ 5(\delta(f+900)+\delta(f-900)) + 60(\delta(f+1000)+\delta(f-1000))$$
$$+ 5(\delta(f+1100)+\delta(f-1100)) + 10(\delta(f+1200)+\delta(f-1200))$$



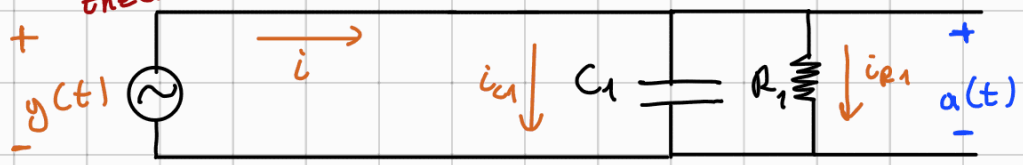
Envelope detector



$$\tau_1 = R_1 \cdot C_1$$

$$\frac{1}{f_c} \ll \tau_1 \ll \frac{1}{B}$$

When $y(t) > 0$ the diode will be short circuit



$$V_g = V_{C_1} = V_{R_1}$$

$$i = i_{C_1} + i_{R_1}$$

$$i_{R_1} = \frac{v_y}{R}, \quad i_{C_1} = C_1 \frac{dv_y}{dt}$$

Applying KCL to parallel RC circuit

$$i_{R_1} + i_{C_1} = 0$$

$$\frac{v_y}{R_1} + C_1 \frac{dv_y}{dt} = 0$$

So when $y(t)$ is less than the previous $y(t)$ value its mean $y(t) = V_g$ less than the capacitor voltage the capacitor will be discharged

↓

and $V_g(t) = V_{C_1}(t) = A e^{-\frac{(t-t_0)}{R_1 C_1}} = V_{C_1}(t_0) e^{-\frac{(t-t_0)}{R_1 C_1}}$

↓ the first voltage when discharge start //

Otherwise $g(t)$ is the output.

The output $a(t)$ will have a dc component anol ripples.

To filter the dc component, we can add a high pass filter with 50 hz cutoff freqency. So the dc component will removed.

Than to filter the ripples we can add a low pass filter, the cutoff frequency can choosed a little nighter than the most high frequency in $m(t)$ message signal

So finally the system diagram used to envelope detecting is as follow.

